1. Crypto Dashboard with Live Data

This is a project to put your React skills into practice after going through the React 19 course. You need to build a crypto dashboard using **CoinGecko API** (free tier) with TanStack Query for data fetching.

Tech Stack

- **Data Fetching:** TanStack Query (React Query)
- **API:** CoinGecko Crypto Data API: Most Comprehensive & Reliable Crypto Price ...
- **Charts:** Recharts

What You'll Practice

The goal is to build a clone of this dashboard table with the following functionalities

Highlights | Categories | Made in USA | DRC-20 | CNY Stablecoin

Coin		Price	1h	24h	7d	24h Volume	Market Cap
Bitcoin BTC	Buy	$109,606	0.4%	1.6%	1.9%	$68,967,347,786	$2,183,145,661,754
Ethereum ETH	Buy	$3,841.29	0.2%	0.8%	5.4%	$39,152,144,152	$463,193,230,688
Tether USDT	Buy	$1.00	0.0%	0.0%	0.0%	$121,916,580,656	$182,575,678,801
BNB BNB	Buy	$1,086.99	0.3%	1.6%	8.5%	$3,357,333,327	$151,261,357,385
XRP XRP	Buy	$2.40	0.0%	0.7%	2.5%	$3,573,017,572	$143,705,732,899
Solana SOL	Buy	$190.02	0.0%	4.1%	3.9%	$7,462,705,867	$103,652,030,018
USDC USDC	Buy	$0.9998	0.0%	0.0%	0.0%	$12,714,208,727	$76,470,877,212
Lido Staked Ether STETH	Buy	$3,838.66	0.3%	0.9%	5.5%	$84,375,170	$32,519,427,566
TRON TRX	Buy	$0.3204	0.5%	0.1%	0.8%	$746,842,054	$30,354,066,262
Dogecoin DOGE	Buy	$0.194	0.1%	2.4%	3.2%	$2,132,962,004	$29,361,568,904
Cardano ADA	Buy	$0.6389	0.1%	1.1%	6.3%	$1,164,618,920	$23,329,542,294
Wrapped stETH WSTETH		$4,666.32	0.3%	0.8%	5.5%	$25,976,974	$14,996,570,174

Features

Core:

- Display 100 initial coins with price, coin name, and other properties as you see them here
- Filters: price range, market cap, volume, gainers/losers
- Sort by: Price/Coin/Other fields on the table
- Only the table dashboard is the important part of this practice; don't pay attention to the navbar or cards above on the website. These are nice to have, but the core of it is the table of coins and its functionality.

Bonus:

- Real-time updates (60s interval)
- Dark mode toggle (persist preference)
- Add coins to favorites

Practice Topics

While building this dashboard, you'll get the chance to practice the following concepts

Hooks

- `useState` - filters, search, sort state
- `useContext` - theme, favorites, presets
- `useMemo` - memoize filtered/sorted data
- `useCallback` - optimize handlers
- Custom hooks

TanStack Query

- `useQuery` - fetch crypto data
- `refetchInterval` - auto-refresh every 60s
- `staleTime` - cache management
- `enabled` - conditional fetching
- Query invalidation

Context API

- ThemeContext (dark/light mode)
- FavoritesContext (saved coins)

TypeScript

- Properly type all props interfaces
- Type API responses
- Avoid using any type
- Use generics for reusable components
- Strict null checks

Performance

- Memoize filtered/sorted results
- Virtual scrolling for 100+ items
- React.memo for coin cards

Testing

Use Vitest + React Testing Library. Think about the following pieces when writing unit/integration tests

1. **Components**
2. Custom Hooks
3. Context
4. API Services